

CSMFAB000000000113-Vol-15

Atmospheric Ascent: From Ocean Surface to the Ionosphere

Series: CSMFAB000000000113 | Volume: 15 of 29 | June 2026

1. Atmosphere Model: NRLMSISE-00

The NRLMSISE-00 empirical model (Picone et al., 2002) provides density, temperature, and pressure vs altitude. Key regimes for AMSR-UBSCLS:

Altitude (km)	rho (kg/m3)	T (K)	P (Pa)	Mach 1 (m/s)
0 (sea level)	1.225	288	101325	340
5	0.737	256	54019	320
10	0.414	223	26436	299
20	0.0888	217	5475	295
30	0.0184	227	1197	301
50	0.000978	271	79.8	329
80	0.0000186	199	1.05	283
100	5.6e-7	195	0.032	280
200	2.5e-10	855	8.5e-5	—

2. Drag Deceleration Model

2.1 Drag Force on Ascending Spear

$$F_{\text{drag}} = 0.5 * \rho(h) * v^2 * C_d(M) * A_{\text{frontal}}$$

C_d is Mach-number dependent for a finned ogive body (from aerodynamics literature): - Mach 0-0.8: $C_d = 0.25$
- Mach 0.8-1.2 (transonic): C_d peaks at 0.55 (wave drag onset) - Mach 1.2-3: $C_d = 0.45$ (established supersonic)
- Mach 3-8: $C_d = 0.35$ (declining with Mach number) - Mach 8+: $C_d = 0.25$ (hypersonic, blunt-body relation)

2.2 Drag Loss Integral

For a 2 kg, 15 cm diameter spear launched at $v_0=1462$ m/s (Mach 4.3):

Numerical integration of the trajectory (4th-order Runge-Kutta, 0.001 s timestep):

Altitude (km)	Velocity (m/s)	Mach	Decel (m/s ²)
0 (launch)	1462	4.3	-1540
5	1380	4.3	-892
10	1310	4.4	-520
20	1260	4.3	-108
30	1240	4.1	-22
50	1218	3.7	-0.9
80	1210	4.3	-0.006
100 (Karman line)	1207	—	~0

At 100 km altitude, the spear still has $1207/1462 = 82.5\%$ of launch velocity! The atmosphere is nearly transparent above 30 km.

2.3 Total Delta-V Lost to Drag

$$\Delta v_{\text{drag}} = \int_0^{\infty} (F_{\text{drag}} / m) dt \quad 255 \text{ m/s}$$

Drag loss = 17.4% of launch velocity. Acceptable.

3. Gravity Turn Trajectory

3.1 Optimal Launch Angle

For maximum altitude (no second stage): $\theta_{\text{opt}} = 90^\circ$ (vertical) For maximum range: $\theta_{\text{opt}} = 45^\circ$ (vacuum), $\sim 38^\circ$ (with drag) For orbital insertion: $\theta_{\text{opt}} = 42^\circ \pm 5^\circ$ (gravity turn optimization)

Gravity turn dynamics: The spear is launched at θ_0 from vertical, and gravity gradually pitches the trajectory horizontal — naturally achieving low-drag ascent while building horizontal velocity.

$$d(\theta)/dt = g * \cos(\theta) / v$$

For $v=1462 \text{ m/s}$, $\theta_0=48^\circ$ from vertical (42° from horizontal): Initial pitchover rate = $9.81 * \cos(48^\circ) / 1462 = 0.0045 \text{ rad/s} = 0.26^\circ/\text{s}$

Over the 100 km ascent time ($\sim 85 \text{ s}$): total pitch change = 22° Final angle at 100 km: $48^\circ - 22^\circ = 26^\circ$ from vertical (64° from horizontal) This is favorable for second-stage orbital insertion.

4. Mach Regime Transitions and Spear Behavior

4.1 Transonic Region (Mach 0.8-1.2)

Critical for spear structural integrity: - Wave drag spike: C_d peaks at 0.55 (vs 0.25 subsonic) - Buffeting: oscillatory loads from shock-boundary layer interaction

ZrB2-SiC spear response: no fatigue (ceramic — no crack propagation under cyclic stress) KNbO3-BaTiO3 flutter damper: active at all Mach numbers; most effective at transonic (highest flutter risk)

4.2 Hypersonic Region (Mach 8+, $v > 2700 \text{ m/s}$)

For Scale 3-5 spears with high exit velocities: Stagnation temperature at Mach 10 at 20 km altitude:

$$T_{\text{stag}} = T_{\text{static}} * (1 + 0.5 * (\gamma - 1) * M^2) = 217 * (1 + 0.2 * 100) = 4557 \text{ K}$$

ZrB2-SiC melting point: $3245^\circ\text{C} = 3518 \text{ K}$ **At Mach 10 in dense atmosphere ($< 20 \text{ km}$), ZrB2-SiC approaches its limit.**

Mitigation: 1. Launch at steeper angle to exit dense atmosphere quickly 2. HfB2 nose tip insert ($T_{\text{melt}}=3387^\circ\text{C}=3660 \text{ K}$) for reserve margin 3. Radiation cooling from YInMn Blue coating (emissivity 0.91) provides partial relief

5. Karman Line Crossing Probability by Scale

Scale	v_{exit} (m/s)	Atmospheric $v_{\text{at } 80\text{km}}$ (m/s)	80km crossing?	100km crossing?
Nano	340	62	No (max $\sim 3 \text{ km}$)	No
Micro	690	520	No (max 21 km)	No
Small	1380	1180	Yes	Marginal ($v=8 \text{ m/s}$ residual)
Medium	1040	840	Marginal	No

Scale 3 (Small, 5-50 kg) is the Karman-line threshold configuration.

Citations (Vol 15)

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